



Johnson and Johnson Positron Quickstart

Posit, PBC

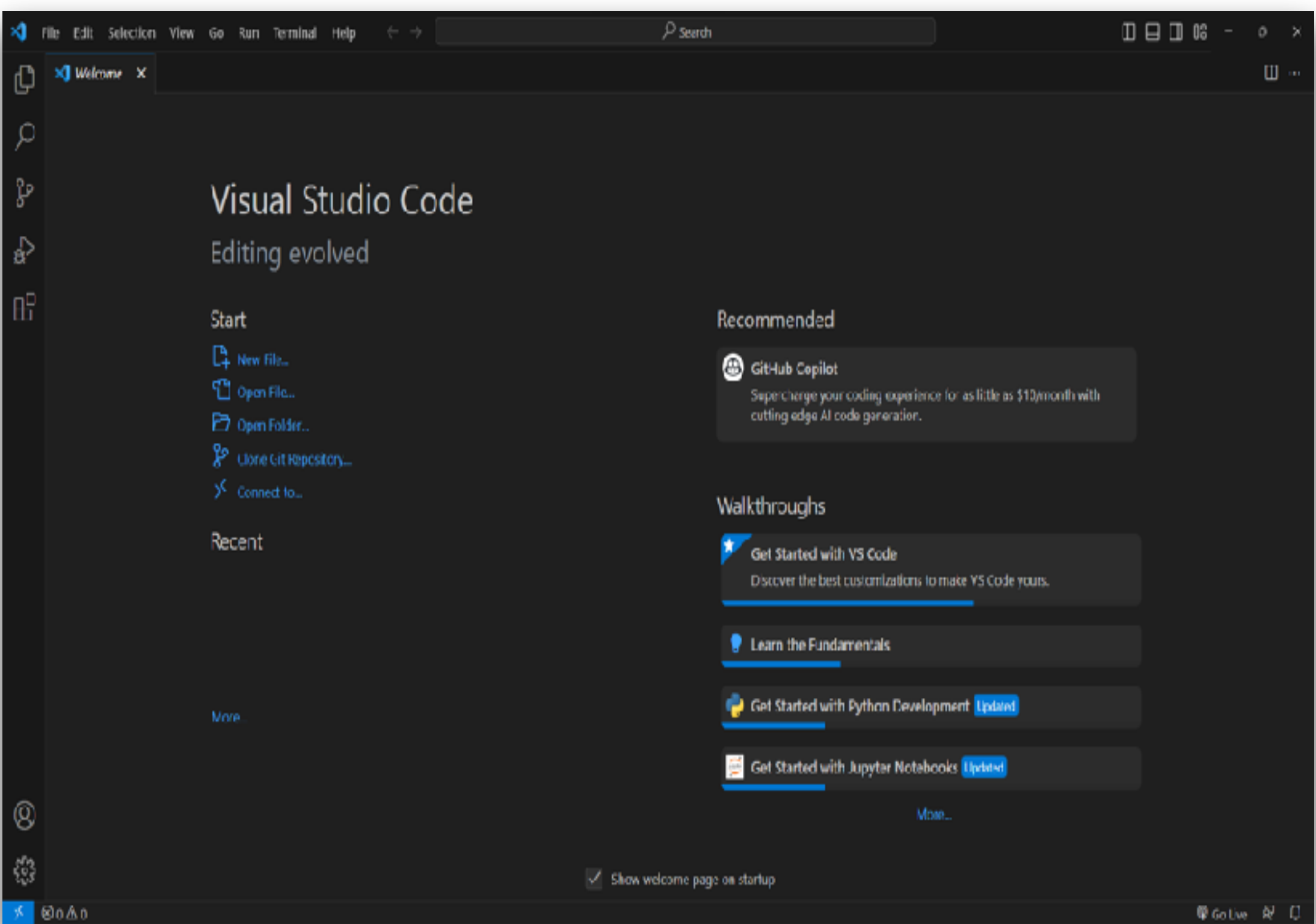
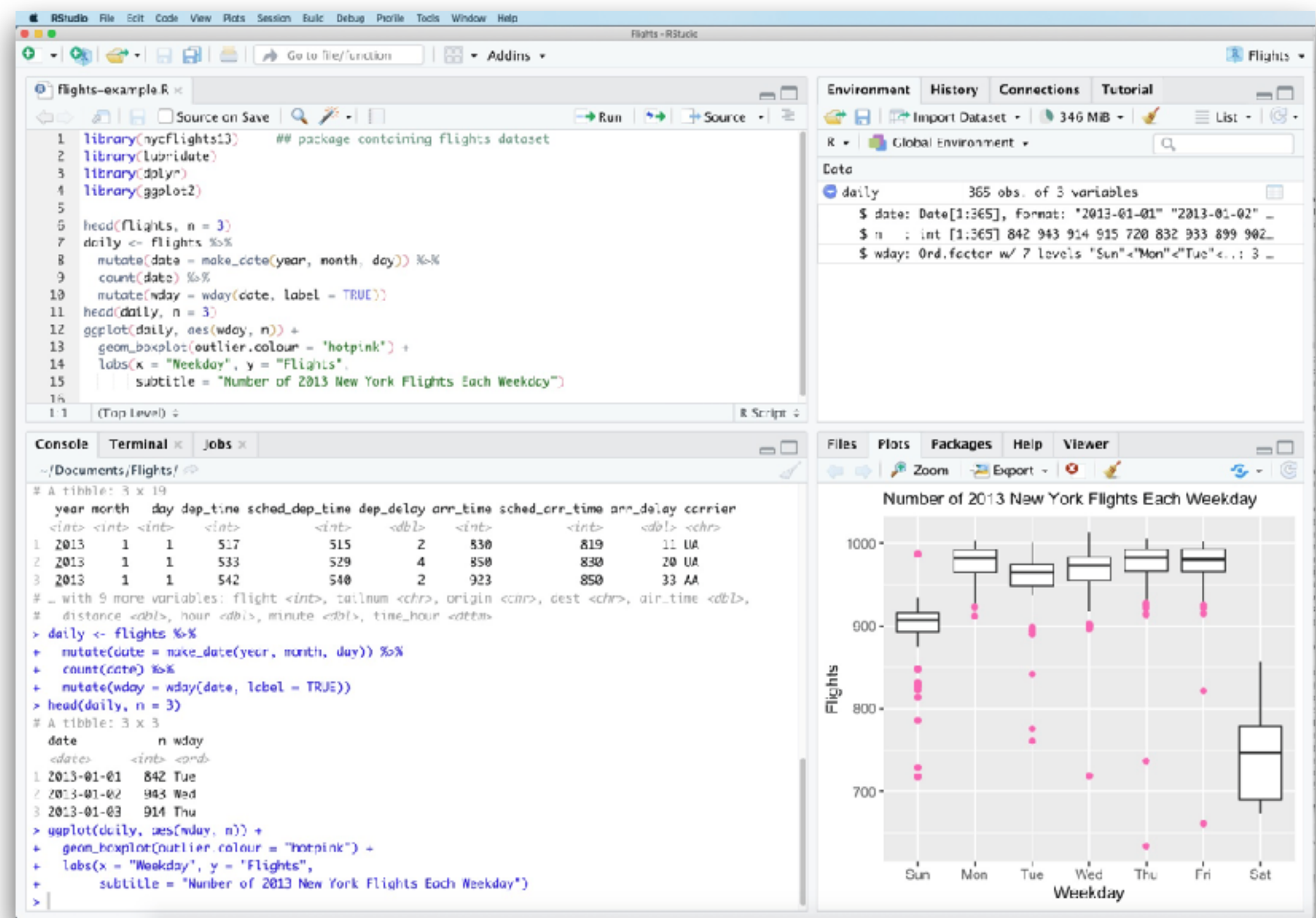
Your Turn

Log into Workbench

Positron is ...

- A next-generation **data science** IDE
- A tool for writing code and exploring data
- Familiar (RStudio-ish) and extensible (soft fork of VS Code)
- Multilingual data science: R and Python (as of today)
- Plots, variables, console, help, viewer, data explorer

Positron is ...



RStudio IDE

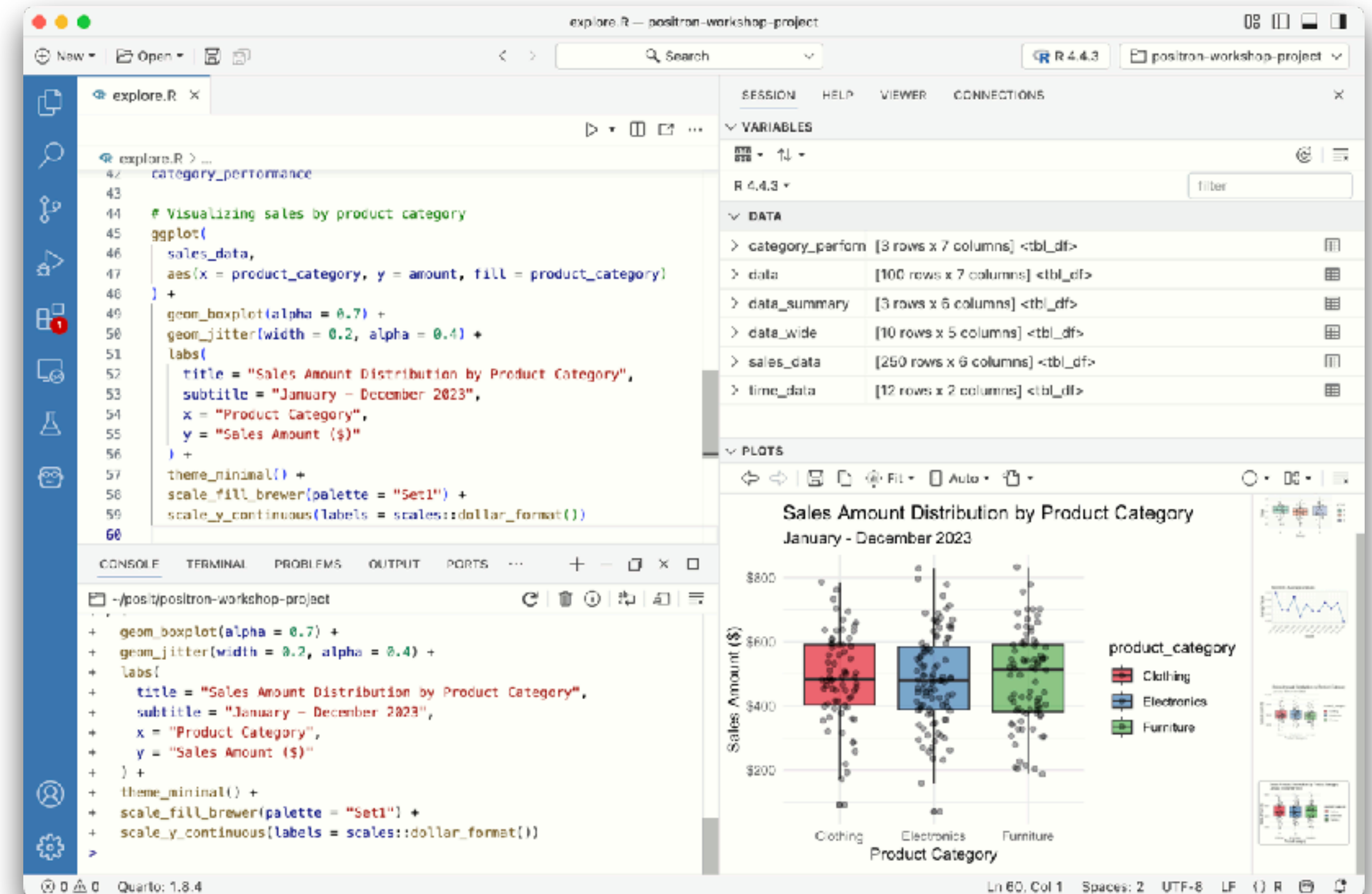
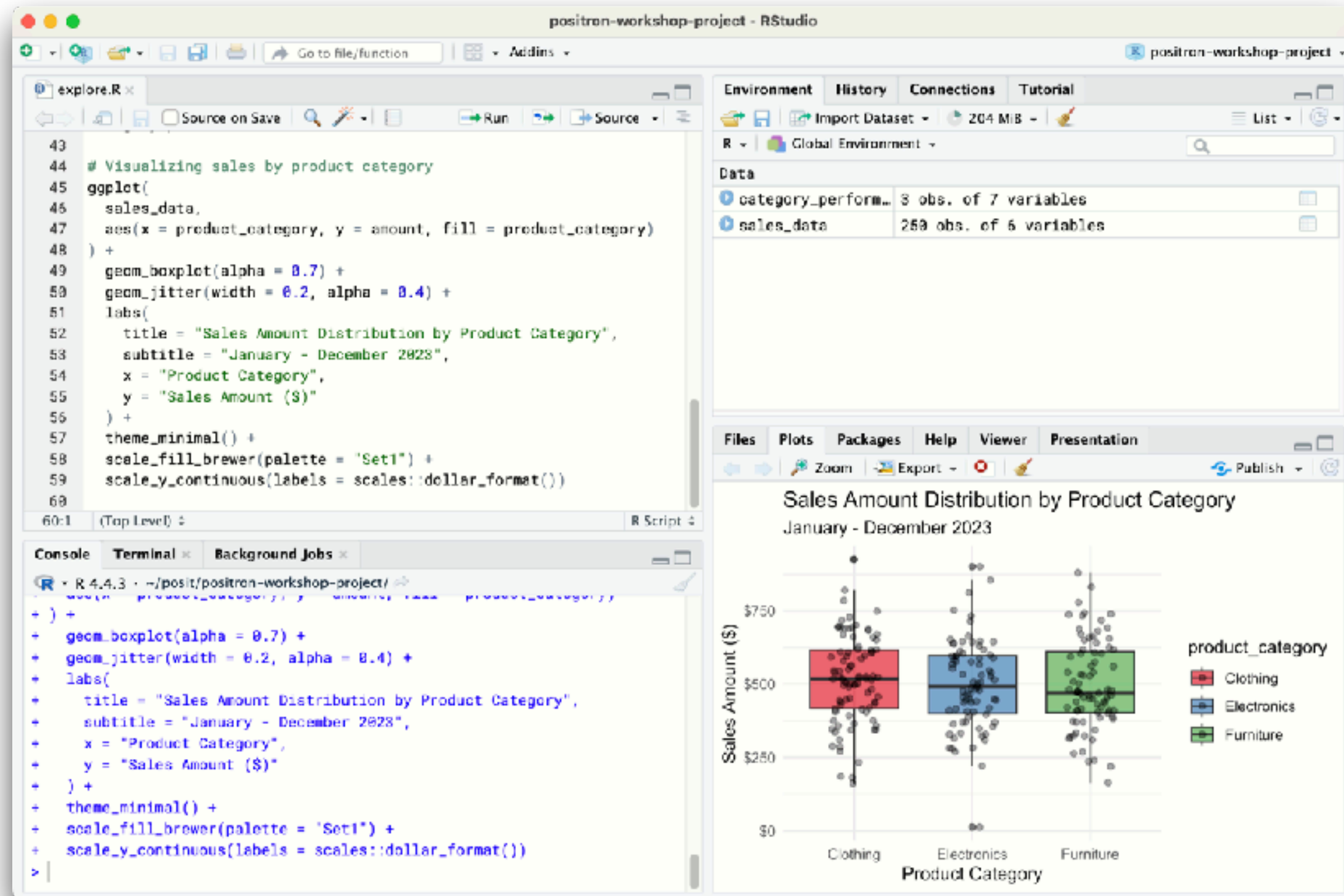
VS Code

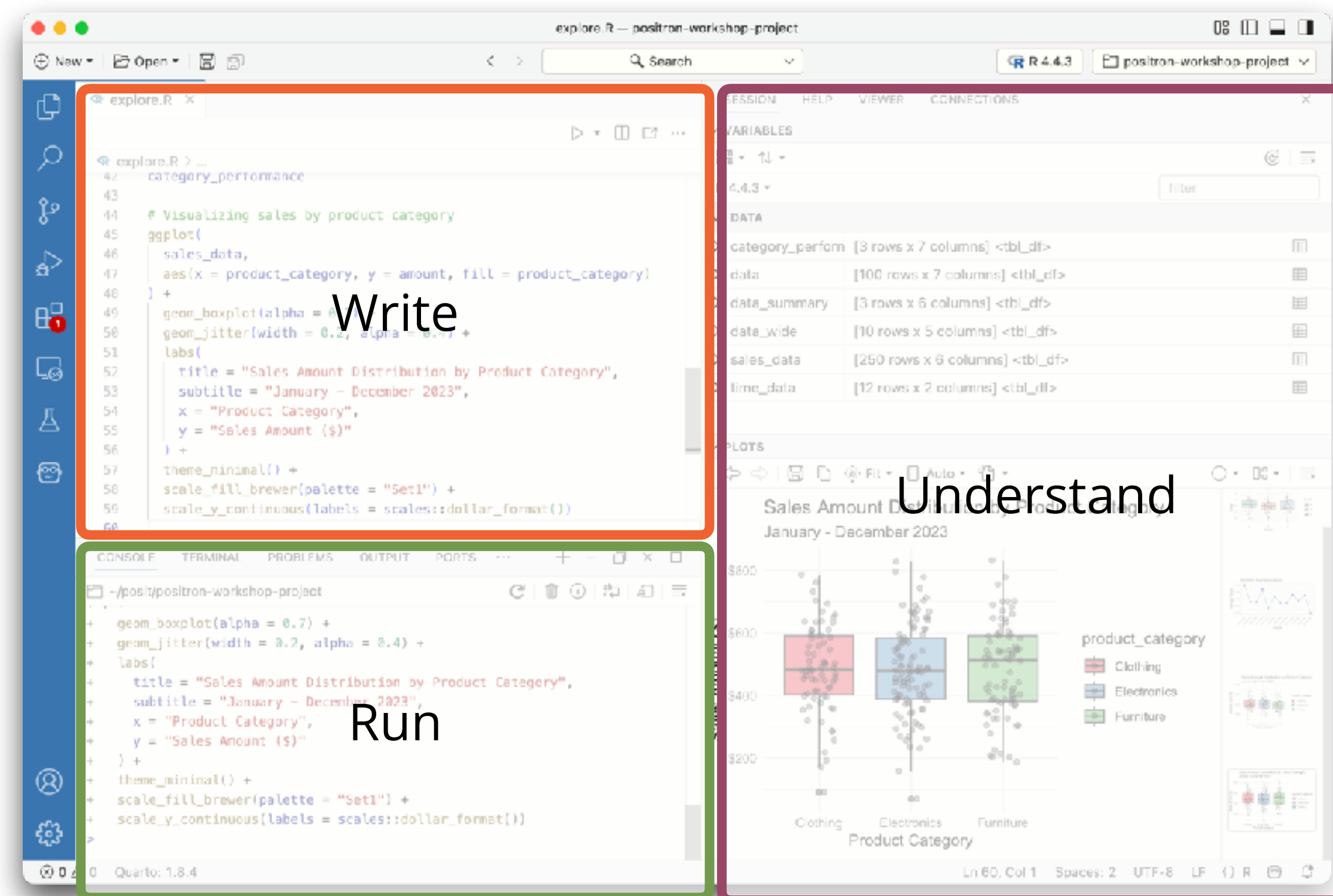
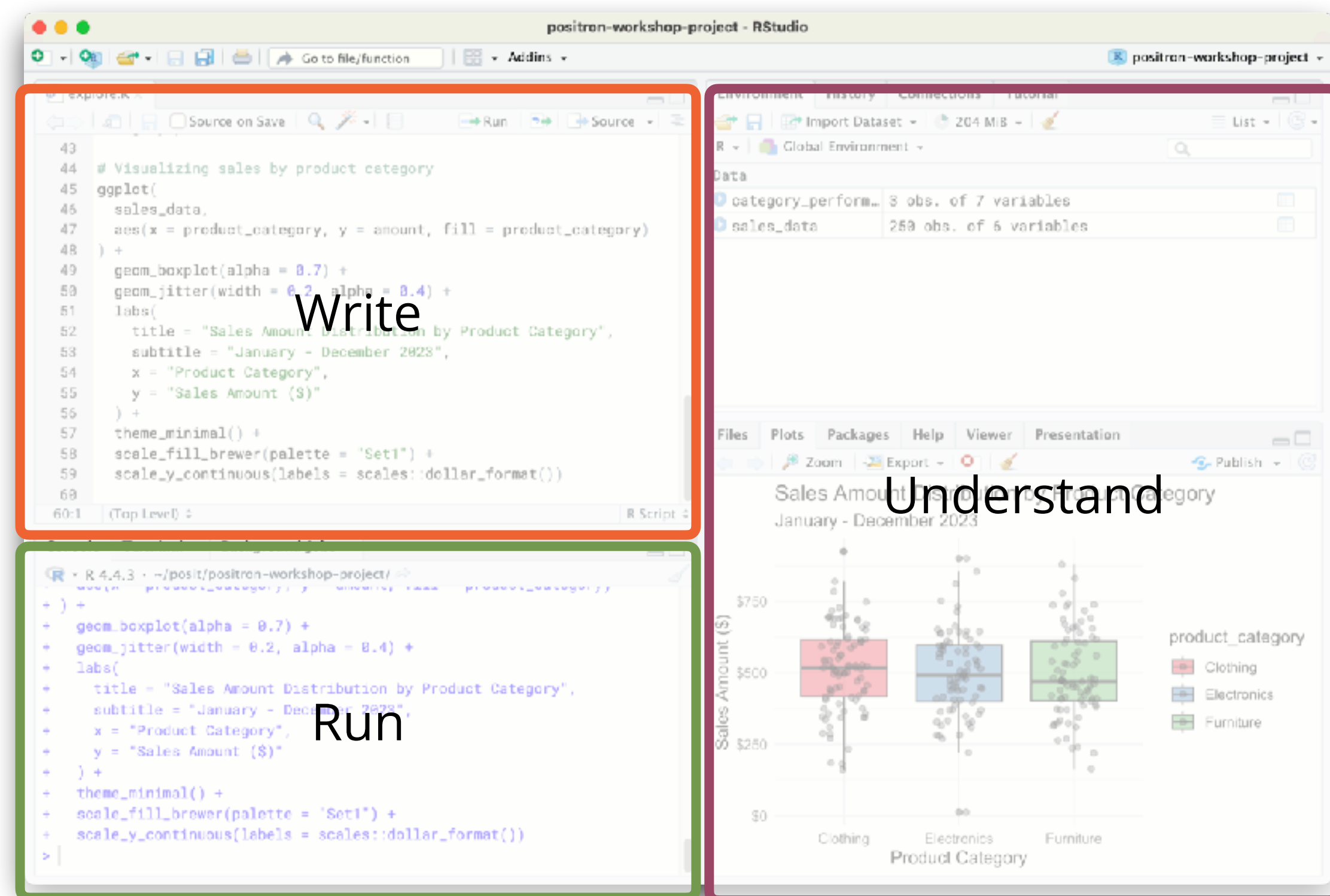
Documentation in two places:

1. positron.posit.co

2. code.visualstudio.com/docs

What's the same?



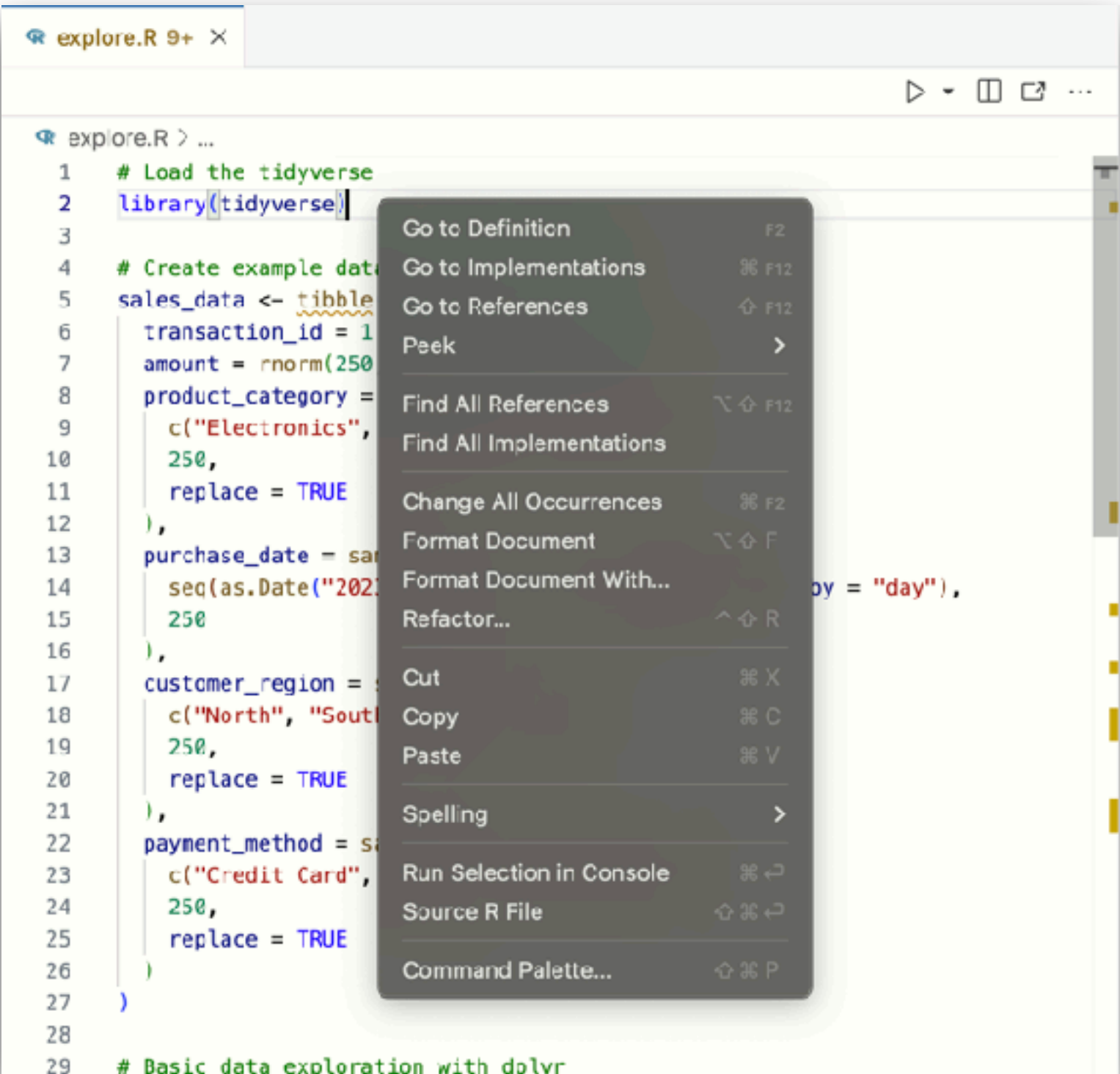


What's different?

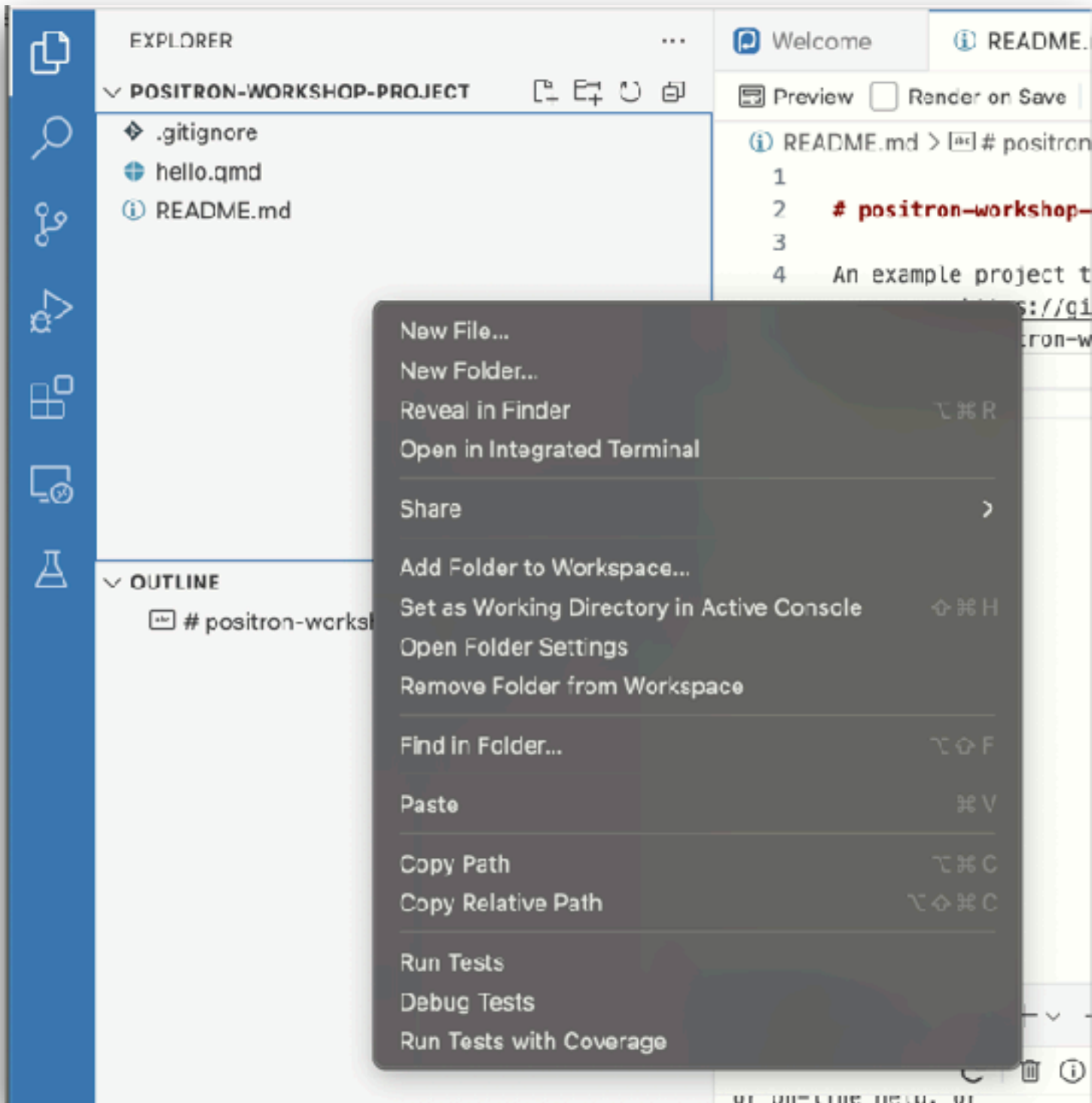
- Projects
 - Languages
 - Versions
 - Restart behavior
 - Command Palette
- Extensions
 - Git pane
 - Air Formatter
 - Snippets
 - Debugging
 - Package Pane
- Settings
 - Layout
 - Keyboard shortcuts
 - Publishing
 - Data viewer
 - ...

Less menus, more right-click

Editor



(File) Explorer



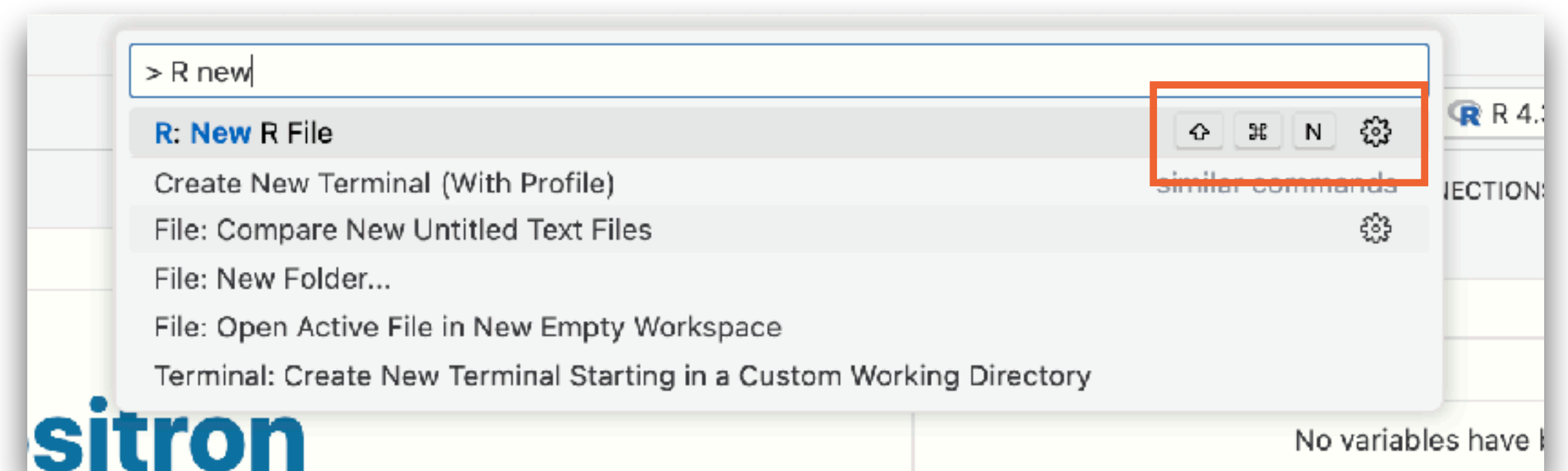
Less clicks, more command palette

Almost everything in a menu or button has a command

Menu: **File > New File > R File**

Command: **R: New R File**

Any command can be given a shortcut





The
command palette
is a core concept in Positron

Ctrl/Cmd + Shift + P

Your Turn

(Ctrl/Cmd + Shift + P)

**Use the Command Palette
to find and open keyboard
Shortcuts**

Learn more

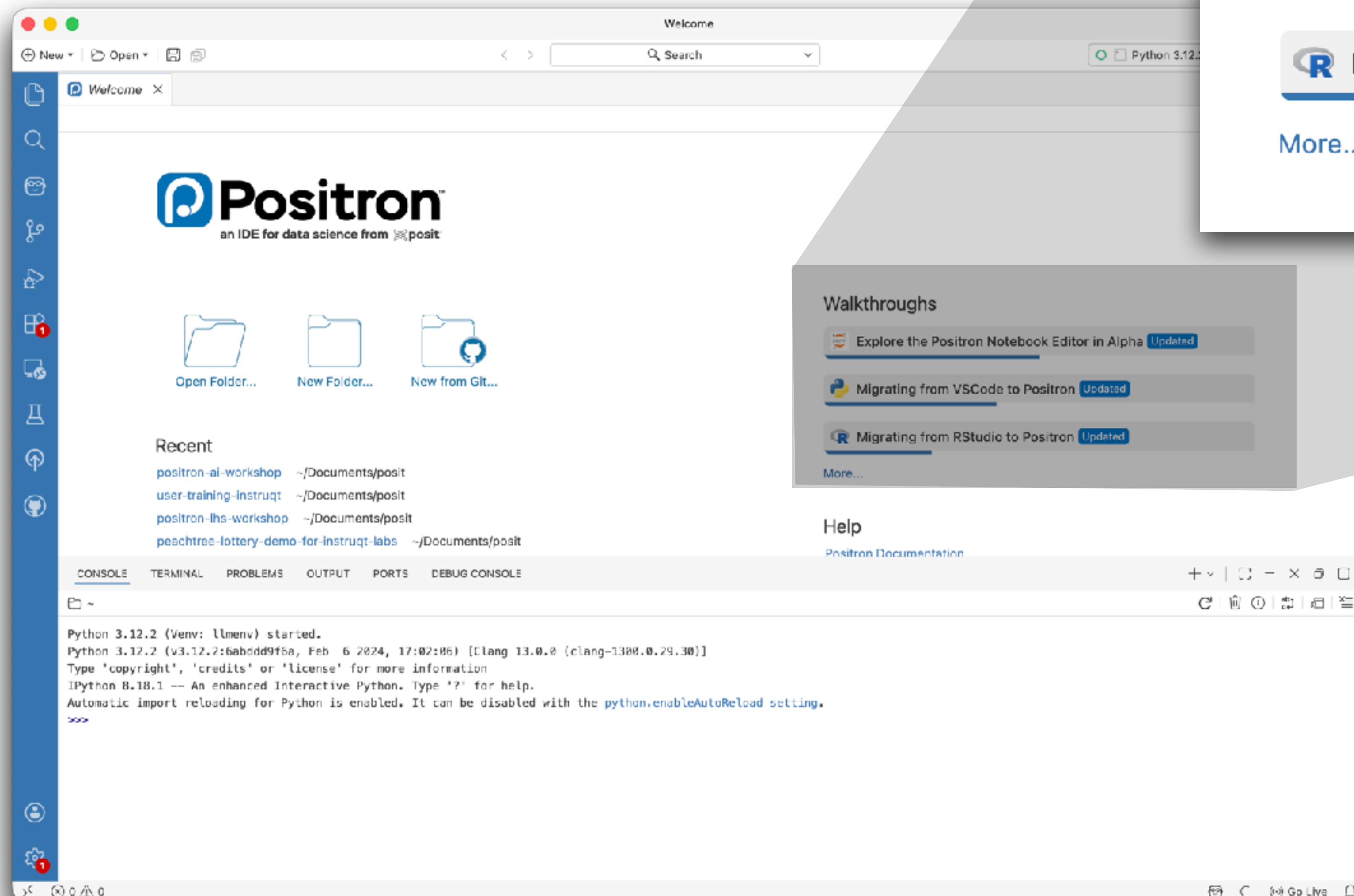
VS Code docs: Keyboard shortcuts for Visual Studio Code

code.visualstudio.com/docs/configure/keybindings

Your Turn

(Ctrl/Cmd + Shift + P)

Use the Command Palette to find
Preferences: Open User Settings.
Then enable RStudio Keybindings.



Walkthroughs



Explore the Positron Notebook Editor in Alpha **Updated**



Migrating from VSCode to Positron **Updated**



Migrating from RStudio to Positron **Updated**

[More...](#)



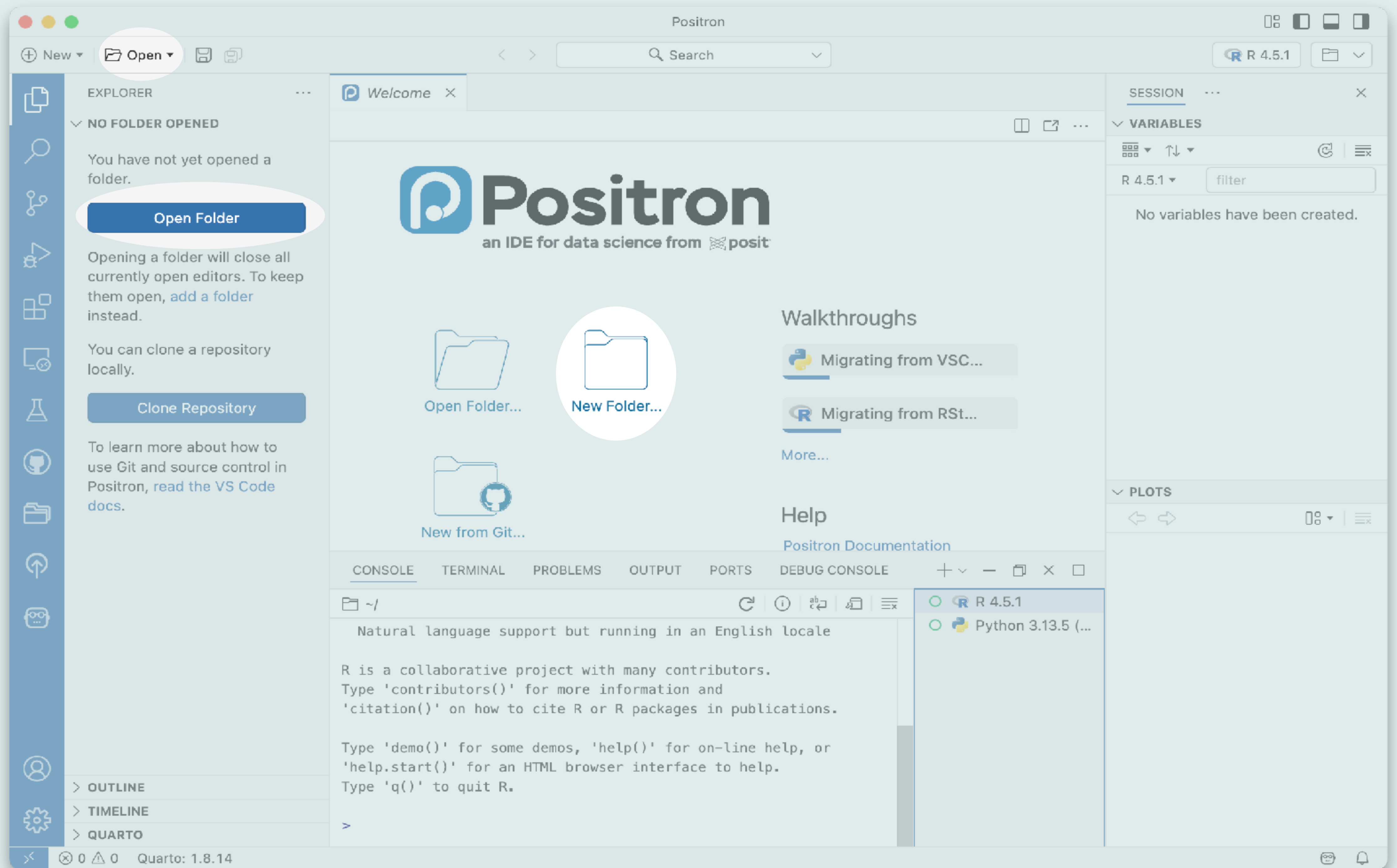
USEFUL

A Workflow



Step 1: Choose a folder

- **Workspace** is a core concept in Positron
- A **workspace** is just a **folder***
- Similar to a **Project** in RStudio
- Pro tip: work in a **folder** most (all?) of the time



What's in a folder?

- Open Folder = dedicated Positron window
- File explorer rooted at the folder.
- Dedicated R/Python process with working directory set to the folder.
- Terminal working directory set to the folder.
- **Can set Workspace settings to override User Settings**

RStudio Projects vs. Positron ???s

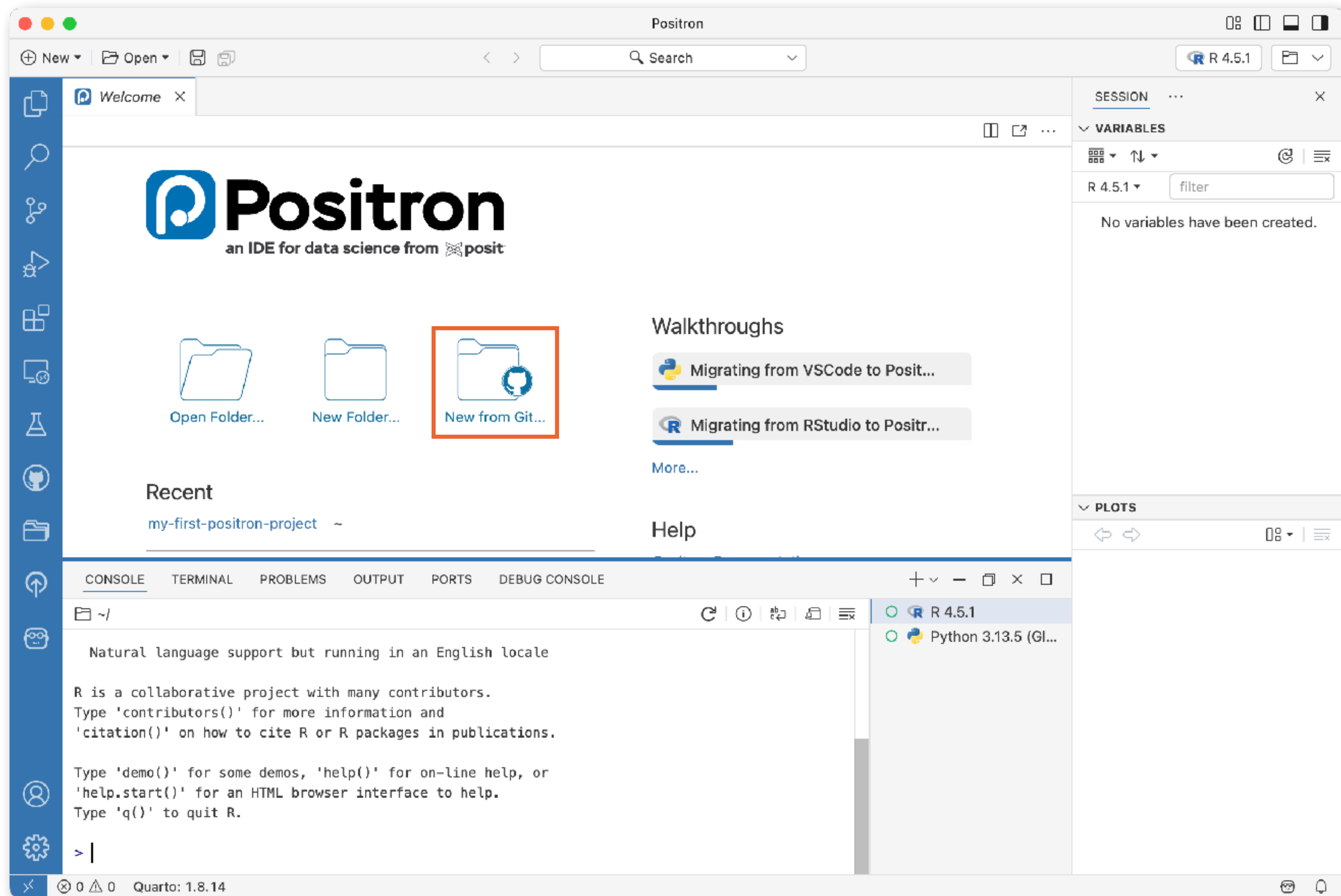
- RStudio places an `.Rproj` file in a folder to designate it as an RStudio Project.
- Used for storing project-level preferences, launching projects, and for identifying the project root (e.g., for `here::here()`).
- Positron doesn't place anything that functions exactly like an Rproj file in every folder.
- Unless you set workspace-level preferences, which prompts the creation of a `.vscode/settings.json` file that stores these preferences.
- Note: In Positron, `here::here()` will find the folder root, typically through finding a `.git` folder.

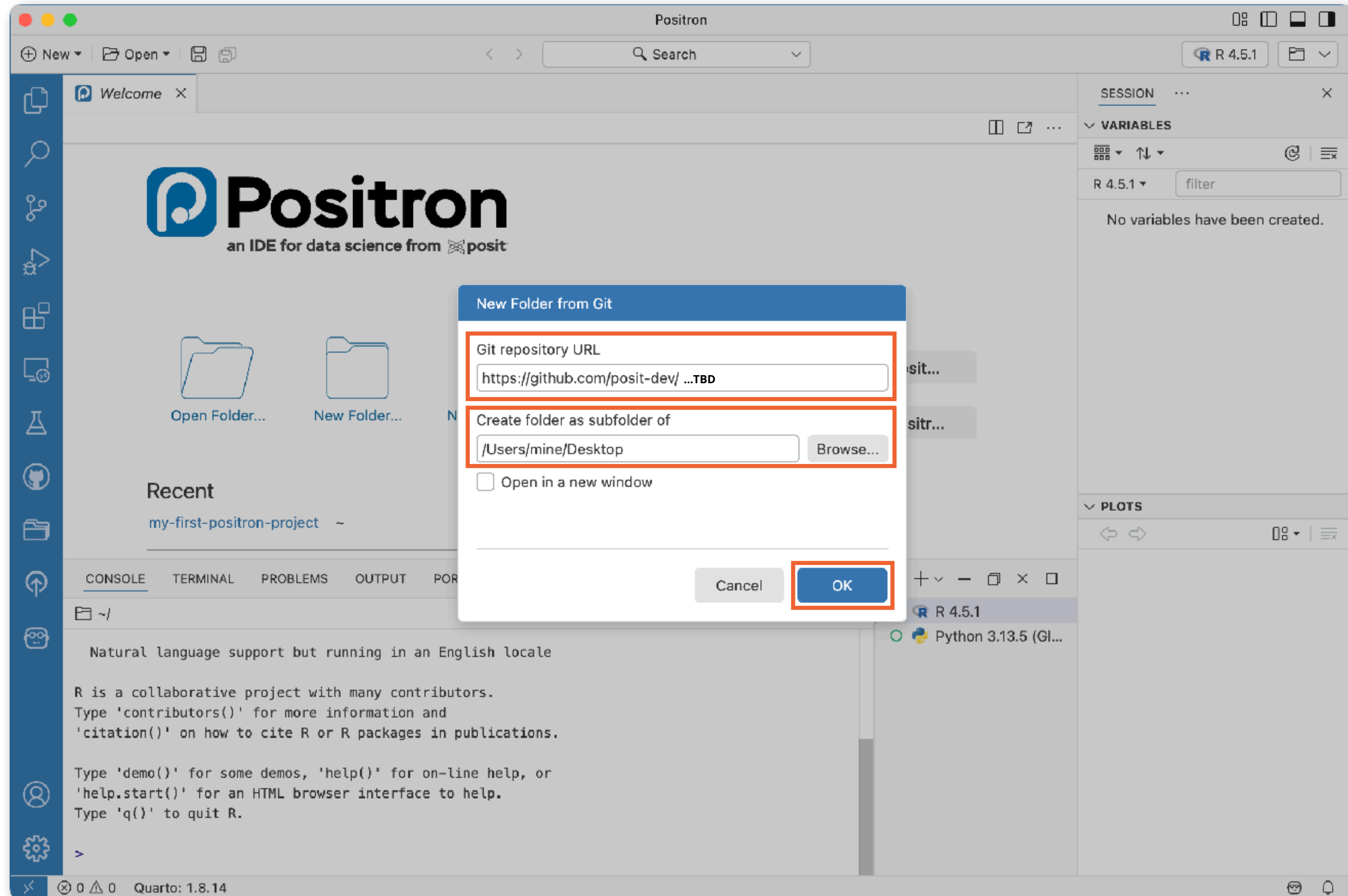


Positron and the `.Rproj` file

- Positron doesn't do anything with the contents of an `.Rproj` file.
- For example, your `.Rproj` file might indicate preferences like using spaces for tabs (`UseSpacesForTab: Yes`) or using line breaks after each sentence in Quarto or R Markdown files (`MarkdownWrap: Sentence`).
- We'll discuss expressing some of these as Positron settings in a later module.
- For Quarto projects, some might be best set in `_quarto.yml`.

r4ds.Rproj	
Version:	1.0
RestoreWorkspace:	Default
SaveWorkspace:	Default
AlwaysSaveHistory:	Default
EnableCodeIndexing:	Yes
UseSpacesForTab:	Yes
NumSpacesForTab:	2
Encoding:	UTF-8
RnwWeave:	Sweave
LaTeX:	XeLaTeX
StripTrailingWhitespace:	Yes
BuildType:	None
UseNativePipeOperator:	Yes
MarkdownWrap:	Sentence
MarkdownCanonical:	Yes
SpellingDictionary:	en_US

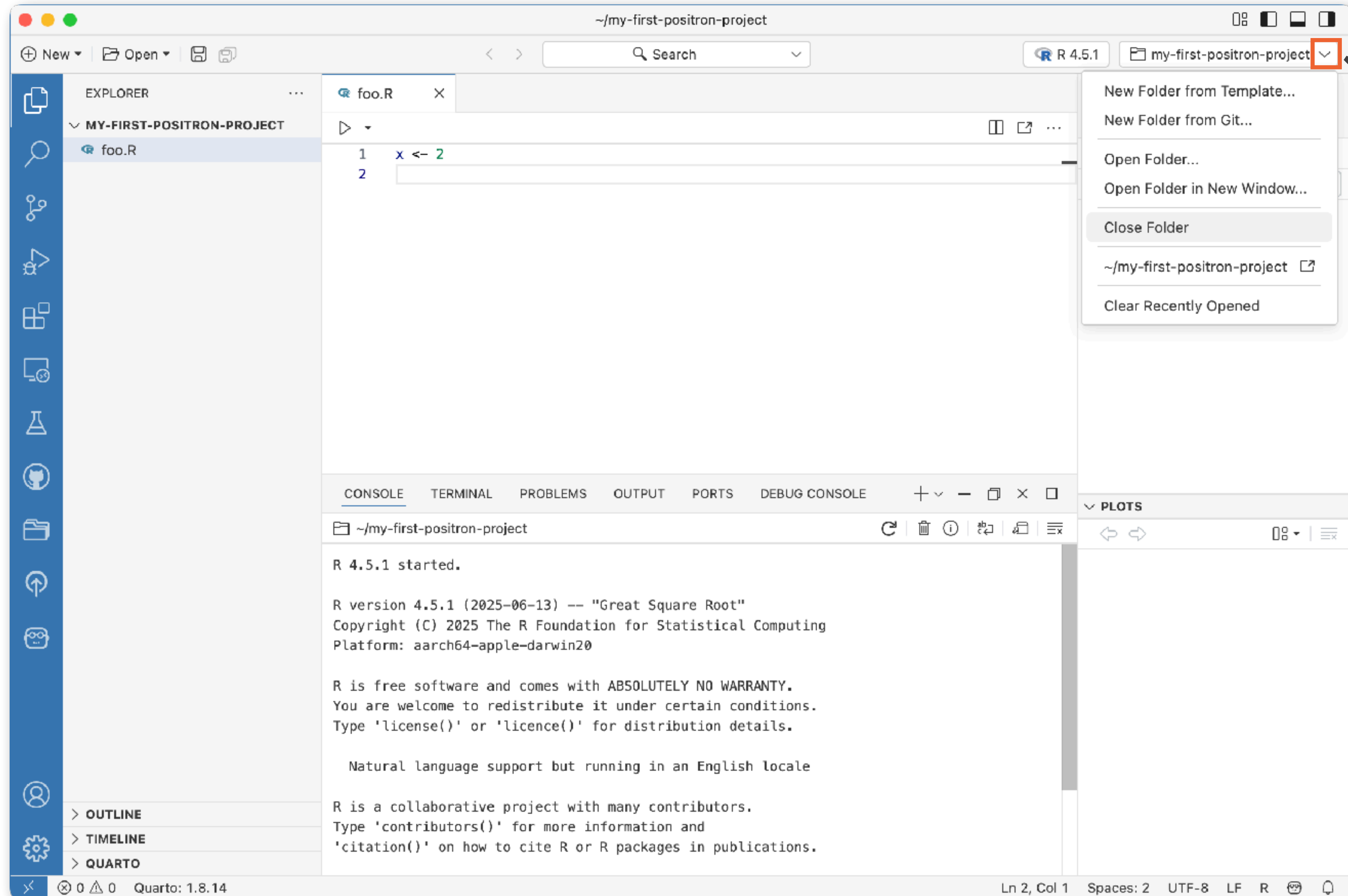




Your Turn

Go to

New > New Folder From Git...



Click to
reveal
menu

Project Manager Extension

(VS Code)

Positron defaults to the "blank slate" lifestyle

```
R --no-save --no-restore-data
```

Step 2: Choose a language and version

R 4.3.3

R 4.4.3

R 4.5.0



New ▾Open ▾Save

EXPLORER

OPEN EDITORS

create.R R

USETHIS

.github

.Rproj.user

.vscode

docs

inst

man

pkgdown

R

addin.R

air.R

author.R

badge.R

block.R

browse.R

ci.R

citation.R

code-of-conduct.R

course.R

coverage.R

cpp11.R

cran.R

create.R

data-table.R

data.R

description.R

directory.R

OUTLINE

TIMELINE

create.R

R > create.R

1 #' Create a package or project

2 #'

3 #' @description

4 #' These functions create an R project:

5 #' * `create\_package()` creates an R package

6 #' * `create\_project()` creates a non-package project, i.e. a data analysis

7 #' project

8 #'

9 #' Both functions can be called on an existing project; you will be asked before

10 #' any existing files are changed.

11 #'

12 #' @inheritParams use_description

13 #' @param fields A named list of fields to add to `DESCRIPTION`, potentially

CONSOLE

TERMINAL

PROBLEMS

OUTPUT

PORTS

DEBUG CONSOLE

~/rrr/usethis

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> devtools::load_all()

i Loading usethis

>

main*

usethis

Quarto: 1.7.31

Ln 1, Col 1

Spaces: 2

UTF-8

LF

R

Select Interpreter Session

R 4.4.2 Currently Selected
/Library/Frameworks/R.framework/Versions/4.4-arm64/Resource...

Active Interpreter Sessions

New Interpreter Session...

R 4.4.2

usethis ▾

SESSION

HELP

...

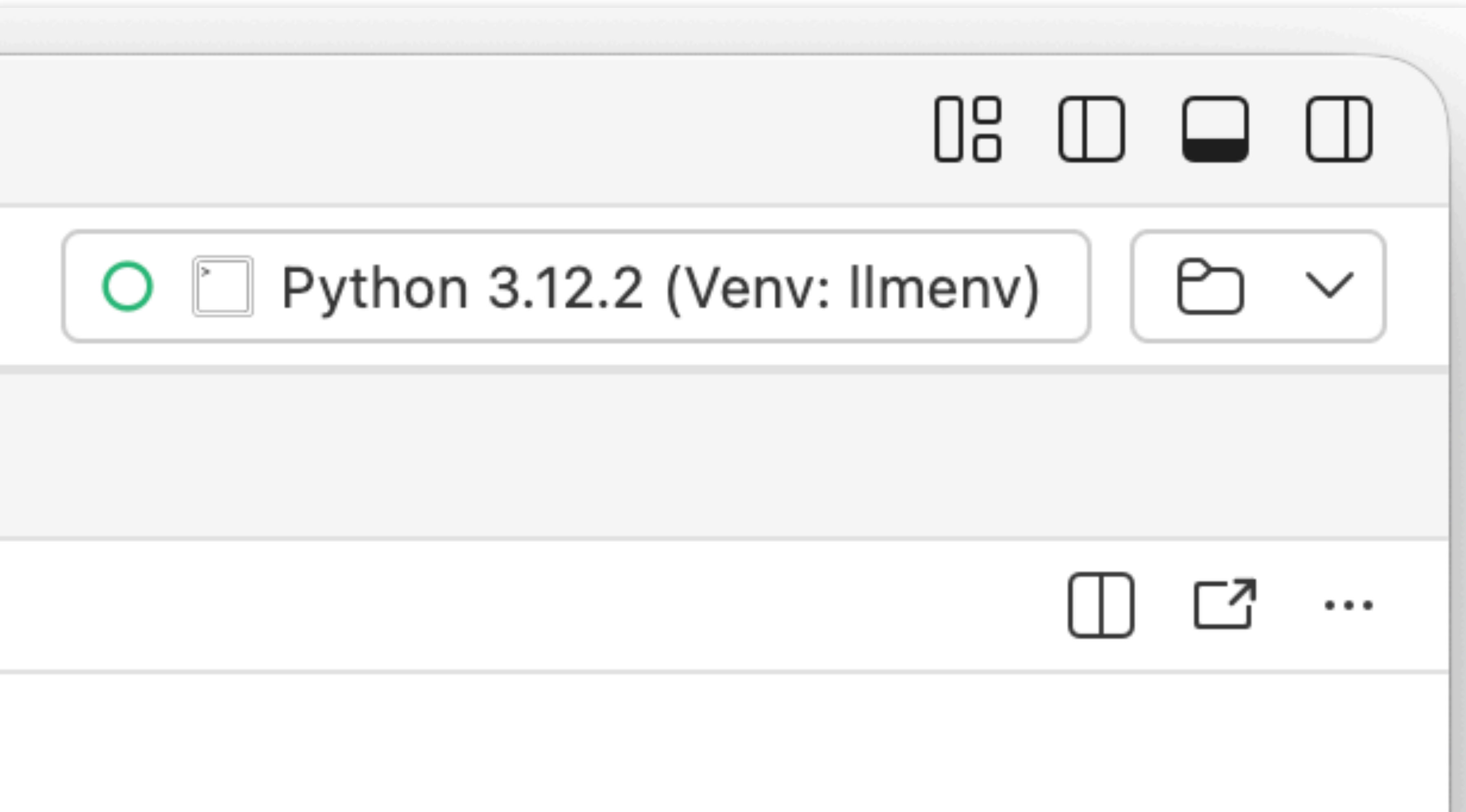
×

The image shows the RStudio application window. A modal dialog titled "Start New Interpreter Session" is open in the center. The dialog has a search bar at the top and a list of interpreters below. The list includes Python 3.11.11 (Pyenv), Python 3.10.16 (Pyenv), Python 3.13.3 (Global), Python 3.9.6 (Global), R 4.4.2 (System), R 4.5.0, and R 4.3.3. The R 4.4.2 interpreter is highlighted with a red border. In the background, the RStudio interface is visible. On the left is the Explorer pane showing a project named "usethis" with a file named "create.R" selected. The main editor pane shows the content of "create.R", which is a template for a new R package. The console at the bottom shows the prompt ">". The session panel on the right shows the current session is "R 4.4.2" and no variables have been created.

Your Turn

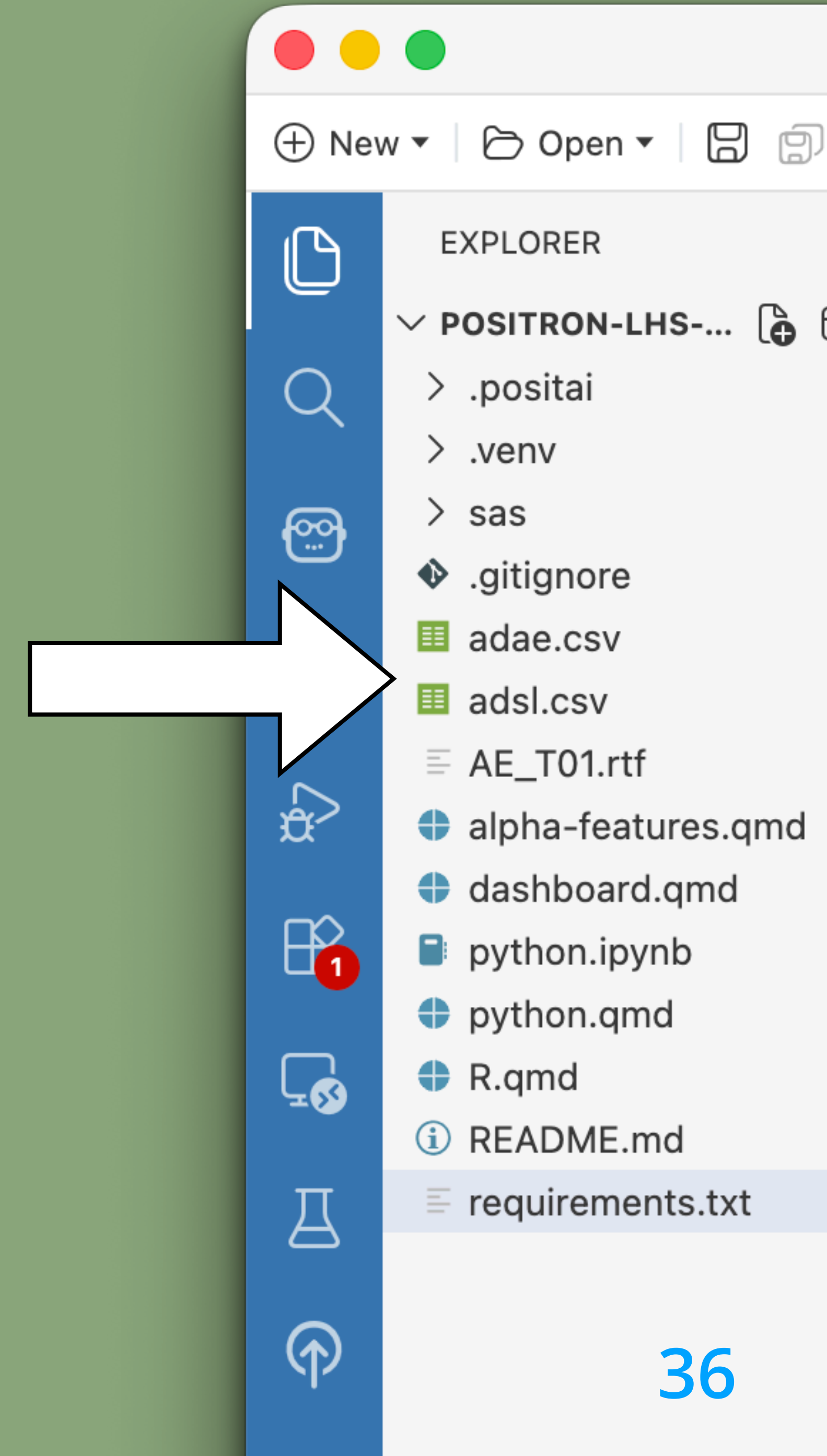
**Start a session with the
interpreter you want to
work with today**

Layouts



Your Turn

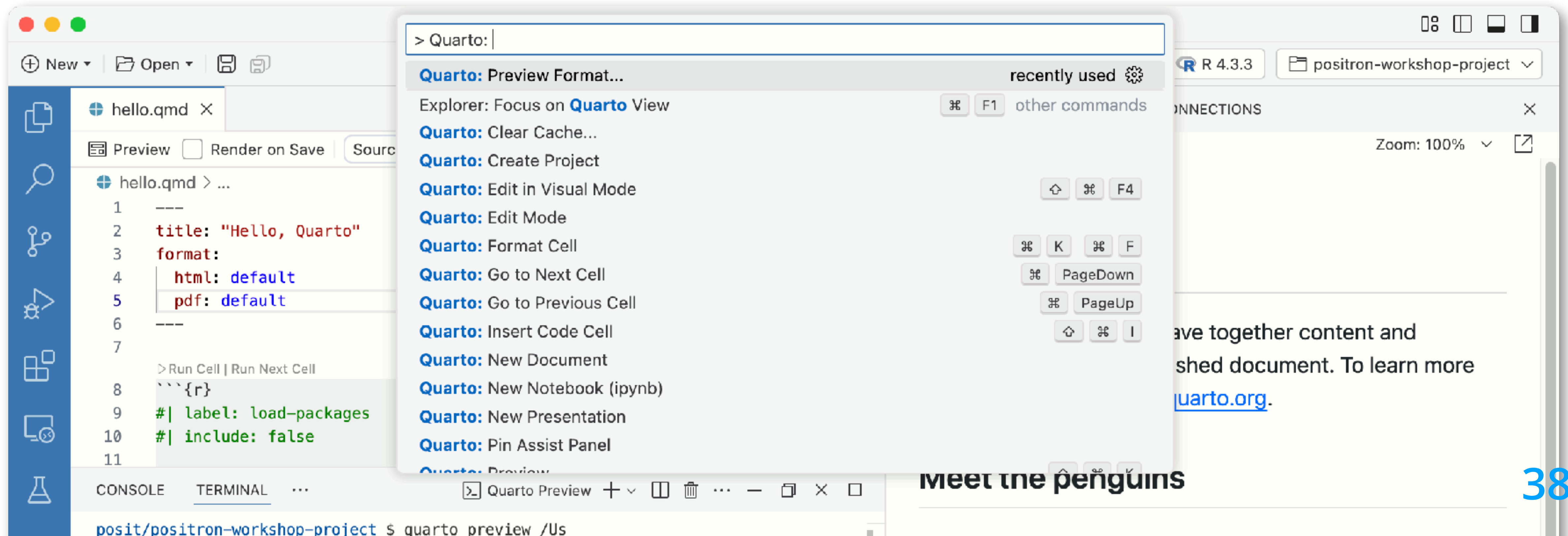
Click on `adae.csv`
and `adsl.csv`. What
can you learn from
the data viewer?



Quarto

Positron is Quarto-ready out-of-the-box

- Bundles Quarto CLI (or uses Quarto it finds on your path)
- Bundles Quarto VS Code Extension



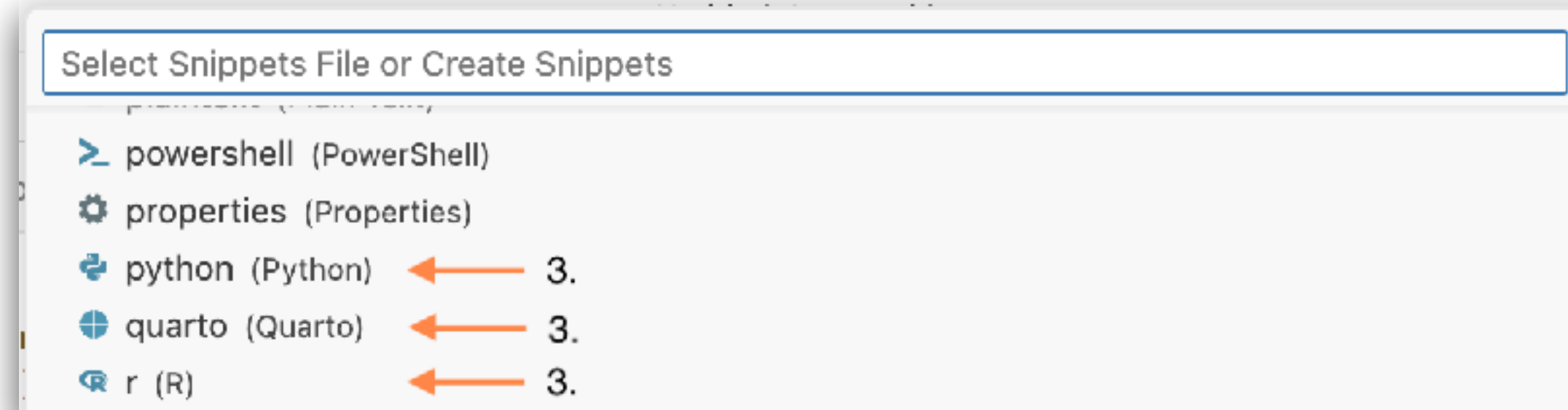
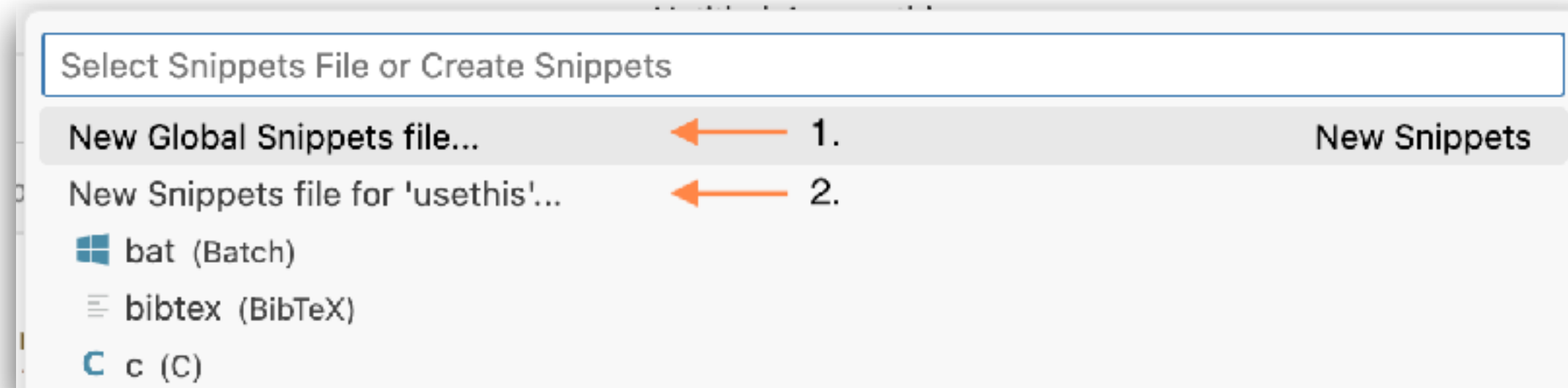
Jupyter Notebooks

When coding, utilize:

- Tab completion
- Snippets
- Help pages

How to configure your own snippets

Command palette: *Snippets: Configure Snippets*

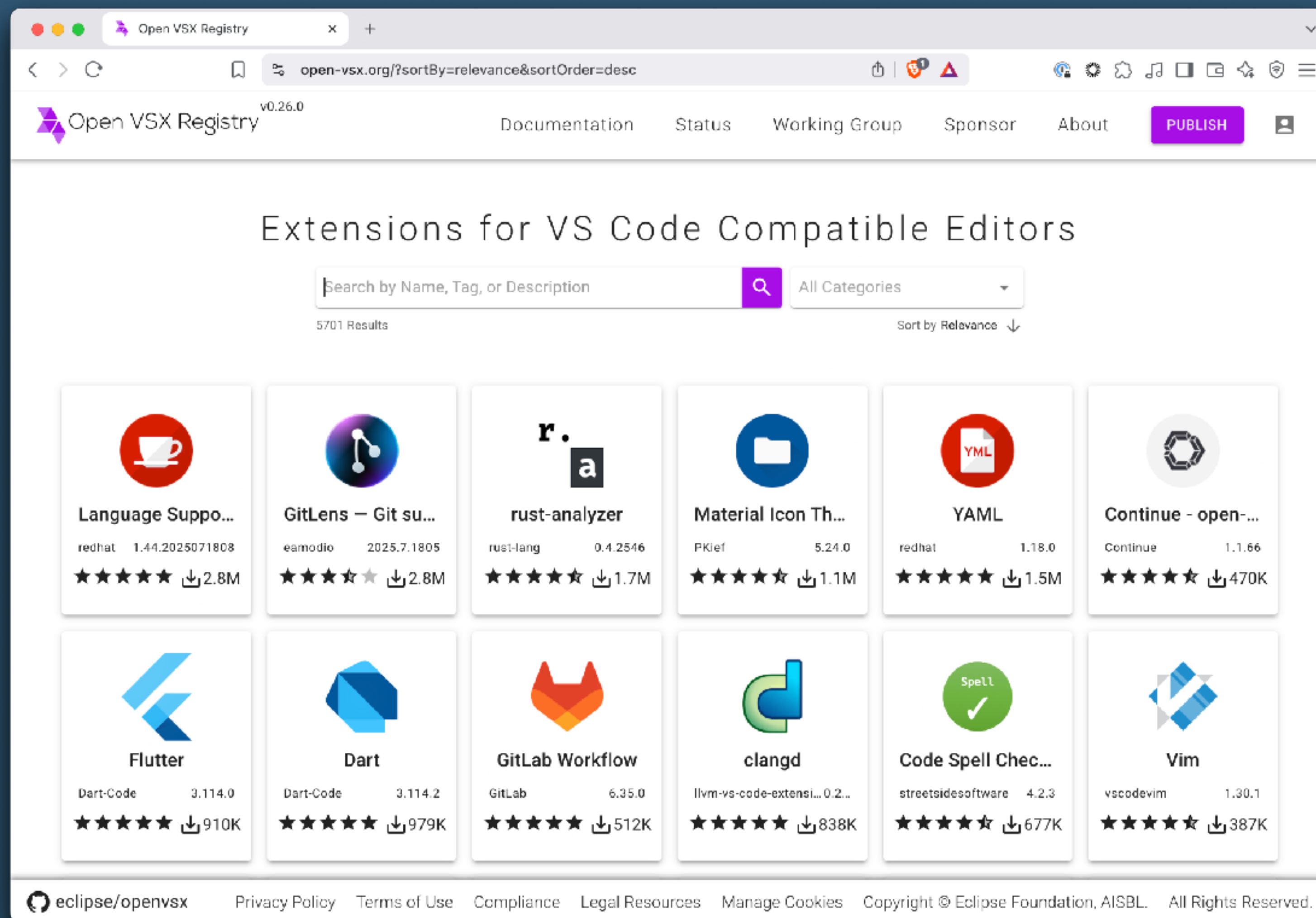


1. Global Snippets file: User-level. Potentially more than 1 language.
2. Workspace-specific file: Specific to 1 workspace. Potentially more than 1 language.
3. Language-specific file: User-level.

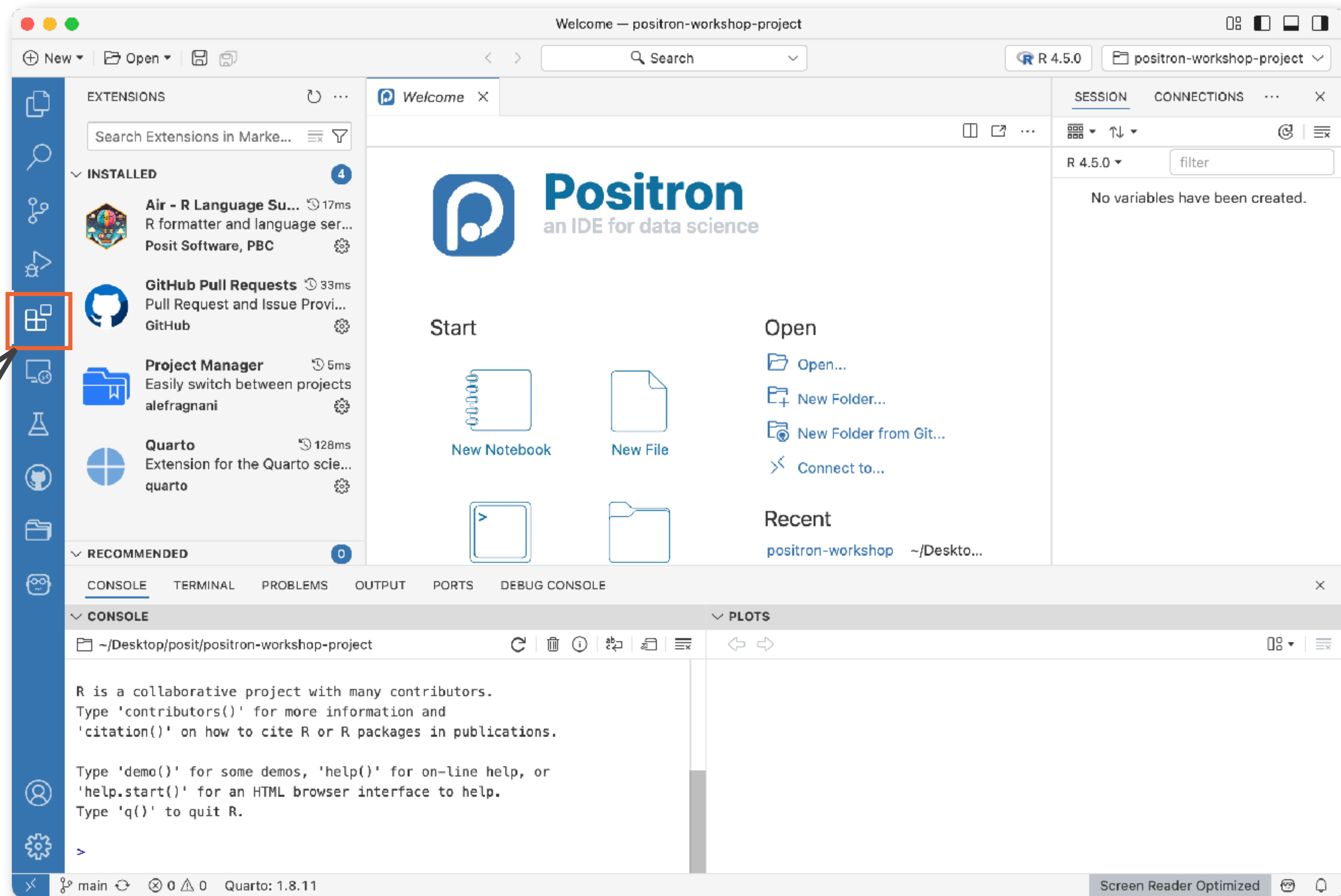
<https://positron.posit.co/r-snippets.html>

Packages...

Extensions



Extensions



Your Turn

Click Preview (Quarto)

Run All (Jupyter)

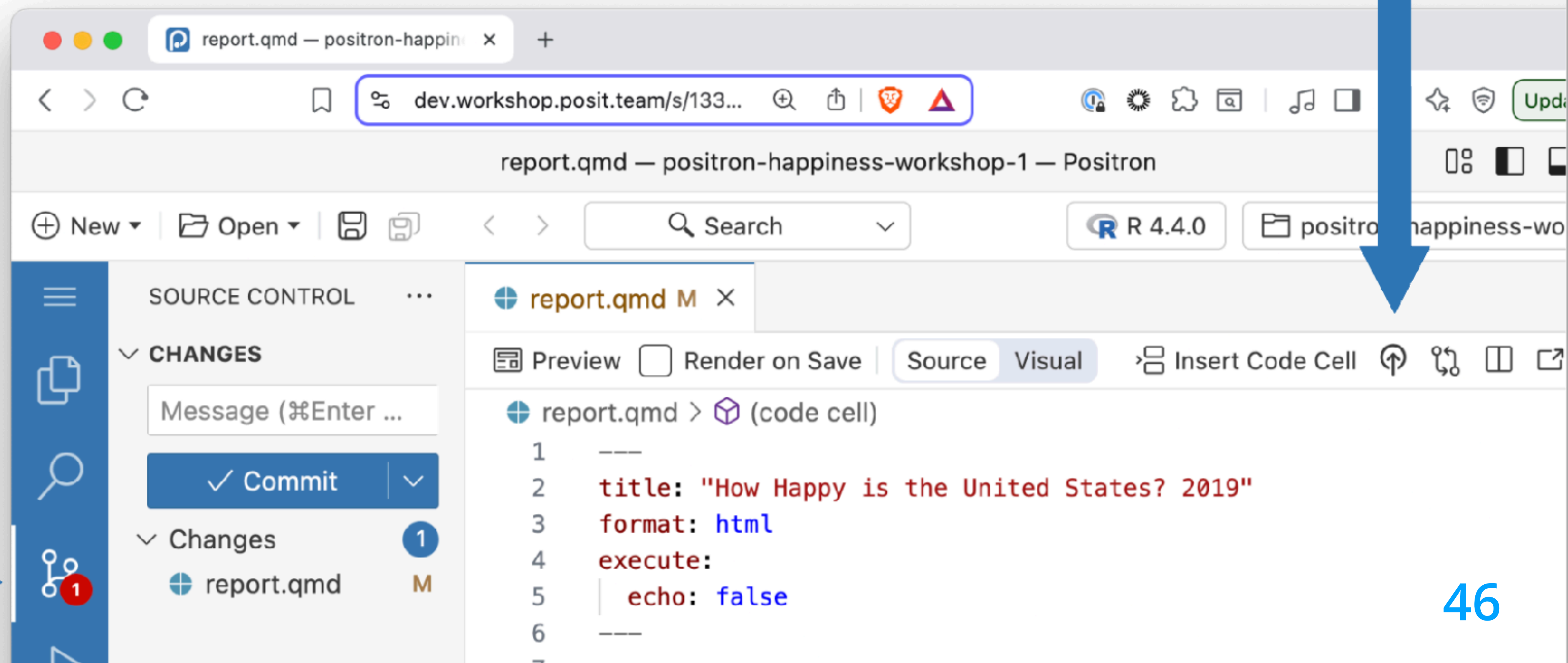
Share where you like



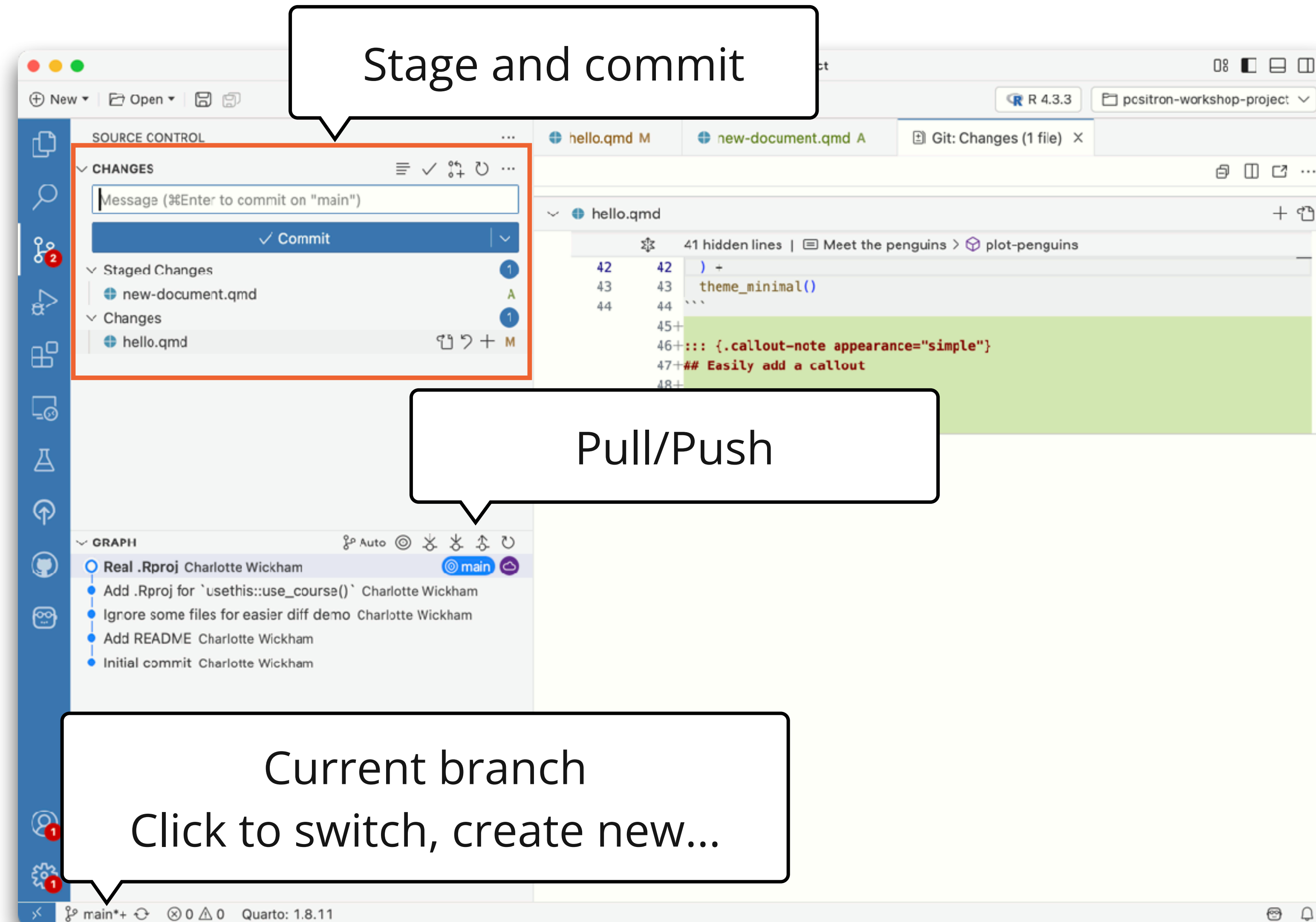
Push button
publishing to
Posit Connect



Push
to source
control for
Git-backed
publishing



Git workflow

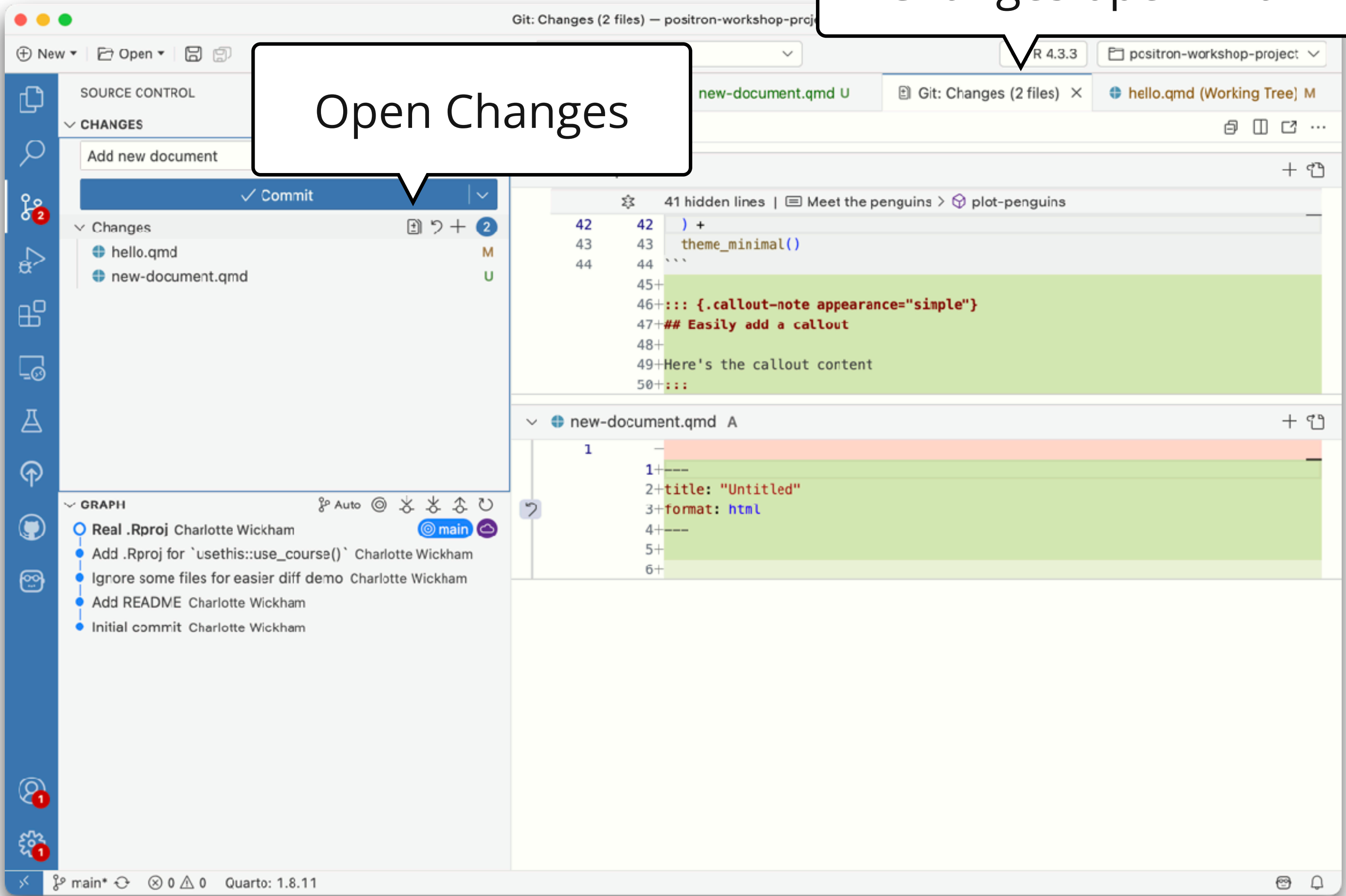


Examine changes

Git Pane

Open Changes

Changes open in an Editor



Software Development

Your Turn

Open functions.R or functions.Py. Source the file. Then call function4().

Built in
debugger



Air formatter for R



posit-dev.github.io/air

- When you install Positron, you get Air "for free" as a bootstrapped extension.
- Air formats your R code based on some opinionated, but also widely-used and agreed-upon, conventions — details at posit-dev.github.io/air/formatter.html.

Air + Positron: practical suggestions

- Air extension ships with Positron and includes the Air binary so Air should just work in Positron.
- You can configure Air to be used at the
 - user level to say “I use Air to format the R code in all of my projects”: by adjusting your user setting (more on this in the next module) or
 - workspace level to say “We use Air to format the R code in this project”: by adjusting your workspace settings (more on this in the next module too) or with `usethis::use_air()`*
- You can also use Air “one-off” from the *Command palette*:
 - *Air: Format Workspace Folder*
 - *Air: Format Document*
 - *Air: Format Selection*
- The Git diff is a great way to see what has changed, so inspect it before committing and pushing!
- There are various ways to disable Air formatting of a specific file, line (or lines in a pipeline), etc.

* Currently requires dev version of usethis

Thank you

Enjoy Positron.